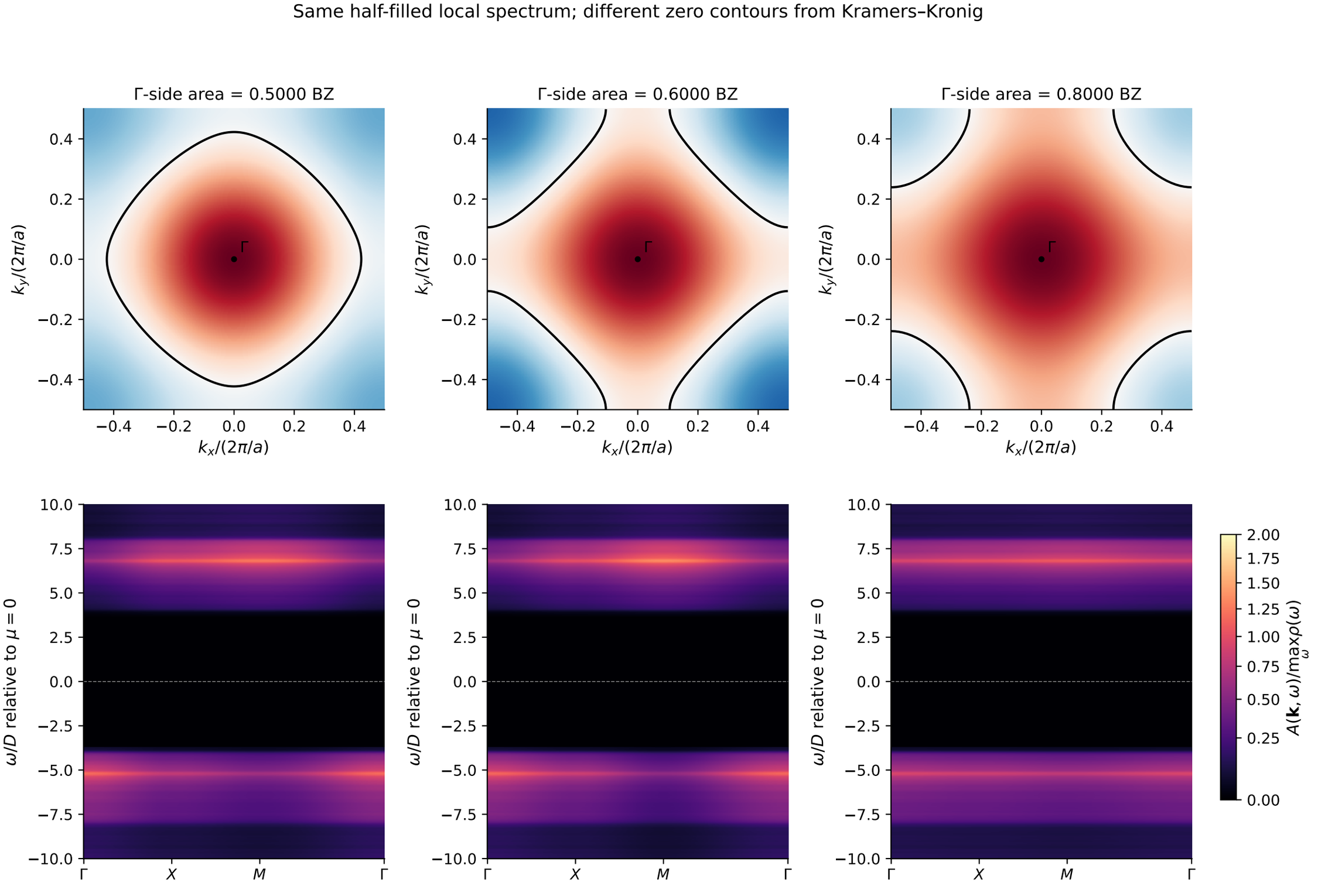

Same half-filled local spectrum; different zero contours from Kramers–Kronig
Γ-side area = 0.5000 BZ
Γ-side area = 0.6000 BZ
Γ-side area = 0.8000 BZ
$k_y/(2\pi/a)$
$k_x/(2\pi/a)$
Γ
ω/D relative to $\mu = 0$
Γ X M Γ
$A(\mathbf{k},\omega)/\max_\omega \rho(\omega)$
Black curves: Re G(k,0) = 0; Im G(k,0) = 0 throughout the gap. Red: positive Re G.
All examples have ν = 1/2, positive spectra, unit spectral sums, C₄ symmetry, and identical momentum averages.